Optimization: Critical Points & Lagrange Multipliers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find all critical points of the function $f(x, y) = x^3 + y^3 - 3xy$.

1. Set the first partial derivatives equal to zero: $f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$. $f_y = 3y^2 - 3x = 0 \Rightarrow x = y^2$. 2. Substitute $y = x^2$ into the second equation: $x = (x^2)^2 = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0$. Real solutions are $x = 0$ and $x = 1$. 3. Determine corresponding y-values:If $x = 0$, $y = 0^2 = 0 \Rightarrow (0, 0)$.If $x = 1$, $y = 1^2 = 1 \Rightarrow (1, 1)$.Thus, the two critical points are $(0, 0)$ and $(1, 1)$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Compute the scalar line integral with respect to arc length $\int_C (x + 2y + 3z)\, ds$, where C is the straight line segment from the origin $(0, 0, 0)$ to $(1, 2, 2)$.

1. Parametrize the line segment: $\mathbf{r}(t) = t\langle 1, 2, 2\rangle = \langle t, 2t, 2t\rangle, \quad 0 \le t \le 1$. 2. Compute differential arc length ds: $\mathbf{r}'(t) = \langle 1, 2, 2\rangle \Rightarrow ||\mathbf{r}'(t)|| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3 \Rightarrow ds = 3\, dt$. 3. Evaluate integrand along C: $x + 2y + 3z = t + 2(2t) + 3(2t) = t + 4t + 6t = 11t$. 4. Integrate.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the volume of the solid inside the sphere $\rho = 4$ and above the cone $\phi = \frac{\pi}{4}$.

$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^4 \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 2\pi\left(\frac{64}{3} - \frac{32\sqrt{2}}{3}\right)$. Final Answer: $2\pi\left(\frac{64}{3} - \frac{32\sqrt{2}}{3}\right)$

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use Green's theorem to evaluate $\oint_C -y^3\, dx + x^3\, dy$, where C is the triangle with vertices $(0,\ 0)$, $(3,\ 0)$, $(0,\ 3)$, oriented counterclockwise.

$\oint_C P\, dx + Q\, dy = \iint_D (Q_x - P_y)\, dA$ with $Q_x - P_y = 3x^2 + 3y^2$. $\int_0^3 \int_0^{3-x} (3x^2 + 3y^2)\, dy\, dx = \frac{81}{2}$.

5 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

For $f(x, y) = x^2 - 6xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx} f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

Final Answer: -24. $f_{xx} = 2$, $f_{yy} = 6$, $f_{xy} = -6$, so $D = 2(6) - (-6)^2 = -24$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Use the method of Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 25$.

Step 1: Set up the Lagrange equations: Constraint $g(x, y) = x^2 + y^2 = 25$. $ablaf = \lambda ablag \Rightarrow \langle 3, 4\rangle = \lambda\langle 2x, 2y\rangle$. This gives: $3 = 2\lambda x \Rightarrow x = \frac{3}{2\lambda}, \quad 4 = 2\lambda y \Rightarrow y = \frac{2}{\lambda}$. Step 2: Substitute into constraint: $\left(\frac{3}{2\lambda}\right)^2 + \left(\frac{2}{\lambda}\right)^2 = 25 \Rightarrow \frac{9}{4\lambda^2} + \frac{16}{4\lambda^2} = 25 \Rightarrow \frac{25}{4\lambda^2} = 25$. .

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find and classify all critical points of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$.

Step 1: Set first partial derivatives equal to zero: $f_x = 6x^2 + y^2 + 10x = 0, \quad f_y = 2xy + 2y = 2y(x + 1) = 0$. Step 2: Solve the system: From $f_y = 0$, either $y = 0$ or $x = -1$.If $y = 0$: $6x^2 + 10x = 2x(3x + 5) = 0 \Rightarrow x = 0$ or $x = -5/3$. Critical points: $(0, 0)$ and $(-5/3, 0)$.If $x = -1$: $6(-1)^2 + y^2 + 10(-1) = 0 \Rightarrow 6 + y^2 - 10 = 0 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2$. Critical points: $(-1, 2)$ and $(-1, -2)$.Step 3: Compute second-order partials.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Multivariable Optimization: Critical Points, Hessians & Lagrange Multipliers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Practice Problem 2: Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y, 2x, (x + y)e^z\rangle$ and C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = y + 1$, oriented counterclockwise when viewed from above.

Compute $abla \times \mathbf{F}$: $abla \times \mathbf{F} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; \partial_x, \partial_y, \partial_z; -y, 2x, (x + y)e^z| = \langle e^z - 0,\ 0 - e^z,\ 2 - (-1)\rangle = \langle e^z, -e^z, 3\rangle$. Choose the planar surface S bounded by C on the plane $z = y + 1$. Here $z = g(x, y) = y + 1$. Upward normal vector: $\mathbf{n}\, dS = \langle -g_x, -g_y, 1\rangle\, dA = \langle 0, -1, 1\rangle\, dA$. Dot product.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Practice Problem 1: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle e^y, xe^y + (z + 1)e^z, (z + 2)e^z\rangle$ along the curve $\mathbf{r}(t) = \langle t, t^2, t^3\rangle$ for $0 \le t \le 1$.

First check if $\mathbf{F}$ is conservative: $P_y = e^y,\ Q_x = e^y; \quad P_z = 0,\ R_x = 0; \quad Q_z = e^z + (z + 1)e^z = (z + 2)e^z,\ R_y = 0.$ Notice $R_y = 0$ while $Q_z = (z + 2)e^z eq 0$. Thus $\mathbf{F}$ is not conservative and the Fundamental Theorem cannot be used directly. Evaluate directly: $\mathbf{r}'(t) = \langle 1, 2t, 3t^2\rangle$. $x = t, y = t^2, z = t^3$. $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = e^{t^2}(1) + \left[te^{t^2} + (t^3 + 1)e^{t^3}\right](2t) + (t^3 + 2)e^{t^3}(3t^2)$. Regro.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Practice Problem 1: Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + \sin(yz),\ y^3 + \cos(xz),\ z^3 + e^{xy} \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

Direct surface integration is prohibitive due to cross-coupling transcendental terms. Compute the divergence: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3 + \sin(yz)) + \frac{\partial}{\partial y}(y^3 + \cos(xz)) + \frac{\partial}{\partial z}(z^3 + e^{xy}) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$. Notice that all transcendental cross-terms vanish identically. Apply the Divergence Theorem: $\iint_{\partial E} \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\, dV$. Convert to spherical coordinates where.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Practice Problem 1: Use Green's Theorem to evaluate $\oint_C (y^3 - \ln(x^2 + 1))\, dx + (x^3 + e^{y^2})\, dy$, where C is the circle $x^2 + y^2 = 4$ oriented counterclockwise.

Identify $P = y^3 - \ln(x^2 + 1)$ and $Q = x^3 + e^{y^2}$. Notice that direct line integration would be intractable due to the transcendental terms. Compute the 2D curl: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3x^2 - 3y^2 = 3(x^2 - y^2)$. Notice that both complicated transcendental terms vanish upon differentiation. Apply Green's Theorem over the disk $D = \{(x, y) \mid x^2 + y^2 \le 4\}$: $\oint_C P\, dx + Q\, dy = \iint_D 3(x^2 - y^2)\, dA$. In polar coordinates: $x^2 - y^2 = r^2 \cos(2\theta)$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Practice Problem 2: Evaluate the line integral $\int_C (2y + \cos x)\, dx + (5x + \sin y)\, dy$ along the upper half of the unit circle $x^2 + y^2 = 1$ from $(1, 0)$ to $(-1, 0)$.

The curve C is open. Close the loop by defining L as the line segment along the x-axis from $(-1, 0)$ to $(1, 0)$. The composite curve $\Gamma = C + L$ forms a positively oriented boundary enclosing the upper half-disk D. By Green's Theorem: $\oint_\Gamma P\, dx + Q\, dy = \iint_D (Q_x - P_y)\, dA = \iint_D (5 - 2)\, dA = 3\,\text{Area}(D) = 3\left(\frac{\pi(1)^2}{2}\right) = \frac{3\pi}{2}$. Along L: $y = 0 \Rightarrow dy = 0$, and x runs from -1 to 1.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Evaluate the integral $\int_0^1 \int_y^1 e^{x^2}\, dx\, dy$ by reversing the order of integration.

Step 1: Sketch and identify the region D: The given bounds are $y \le x \le 1$ for $0 \le y \le 1$. This is the triangle bounded by $x = y$, $x = 1$, and $y = 0$. Step 2: Reverse order to integrate with respect to y first: In the reversed order, $0 \le y \le x$ for $0 \le x \le 1$: $\int_0^1 \int_0^x e^{x^2}\, dy\, dx$. Step 3: Evaluate inner integral: $\int_0^x e^{x^2}\, dy = [y e^{x^2}]_{y=0}^{y=x} = x e^{x^2}$. Step 4: Evaluate outer integral using substitution $u = x^2$, $du = 2x\, dx$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Practice Problem 2: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = abla(x^3y^2 + \sin(xz))$ and C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \le t \le 2\pi$.

Since $\mathbf{F}$ is explicitly given as the gradient of the potential function $f(x, y, z) = x^3y^2 + \sin(xz)$, by the Fundamental Theorem of Line Integrals: $\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{r}(2\pi)) - f(\mathbf{r}(0))$. Initial point: $\mathbf{r}(0) = \langle \cos 0, \sin 0, 0 \rangle = (1, 0, 0)$. Terminal point: $\mathbf{r}(2\pi) = \langle \cos(2\pi), \sin(2\pi), 2\pi \rangle = (1, 0, 2\pi)$. Evaluating f at endpoints: $f(1, 0, 0) = 1^3(0)^2 + \sin(0) = 0$. .

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Double Integrals

Evaluate $\iint_D (x^2 + y^2)\, dA$, where D is the upper half-disk defined by $x^2 + y^2 \le 9$ with $y \ge 0$.

Step 1: Convert region D to polar coordinates: The upper half-disk of radius 3 corresponds to $0 \le r \le 3$ and $0 \le \theta \le \pi$. Step 2: Convert integrand and area element: $x^2 + y^2 = r^2, \quad dA = r\, dr\, d\theta$. Step 3: Set up and evaluate iterated integral: $\iint_D (x^2 + y^2)\, dA = \int_0^{\pi} \int_0^3 r^2 (r\, dr\, d\theta) = \int_0^{\pi} d\theta \cdot \int_0^3 r^3\, dr$. $= \pi \cdot \left[\frac{r^4}{4}\right]_0^3 = \pi \cdot \frac{81}{4} = \mathbf{\frac{81\pi}{4}}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Practice Problem 2: Find the mass of a solid right circular cylinder of radius R and height h whose density at any point is proportional to its distance from the axis of symmetry, $\delta(r, \theta, z) = kr$ for constant $k > 0$.

Using cylindrical coordinates, the cylinder E is bounded by $0 \le r \le R$, $0 \le \theta \le 2\pi$, $0 \le z \le h$. The differential volume is $dV = r\, dz\, dr\, d\theta$. The total mass is: $M = \iiint_E \delta\, dV = \int_0^{2\pi} \int_0^R \int_0^h (kr) \cdot r\, dz\, dr\, d\theta = k\left(\int_0^{2\pi} d\theta\right)\left(\int_0^h dz\right)\left(\int_0^R r^2\, dr\right)$. $M = k(2\pi)(h)\left(\frac{R^3}{3}\right) = \frac{2\pi k h R^3}{3}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Practice Problem 1: Evaluate $\iiint_E (x^2 + y^2 + z^2)\, dV$, where E is the solid ball $x^2 + y^2 + z^2 \le a^2$.

In spherical coordinates, $x^2 + y^2 + z^2 = \rho^2$ and $dV = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$. The ball E is defined by $0 \le \rho \le a$, $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$. $\iiint_E \rho^2 \cdot (\rho^2 \sin\phi)\, d\rho\, d\phi\, d\theta = \left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi} \sin\phi\, d\phi\right)\left(\int_0^a \rho^4\, d\rho\right)$. Evaluating each integral: $2\pi \times 2 \times \frac{a^5}{5} = \frac{4\pi a^5}{5}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Double Integrals

[Double Integrals, Fubini's Theorem, Polar Coordinates & The Change-of-Variables Jacobian · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

[Green's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What condition defines a critical point (x_0, y_0) of a differentiable function $f(x, y)$?

a. $\nabla f(x_0, y_0) = \mathbf{0}$**, meaning both** $f_x(x_0, y_0) = 0$ **and** $f_y(x_0, y_0) = 0$ ✓

b. $f_{xx}(x_0, y_0) = 0$

c. $f(x_0, y_0) = 0$

d. $f_x(x_0, y_0) = f_y(x_0, y_0)$

Critical points occur where the gradient vector is zero (both first-order partials vanish simultaneously) or where a partial derivative fails to exist.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the critical points of $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. $(2, -3)$ ✓

b. $(-2, 3)$

c. $(4, -6)$

d. $(0, 0)$

$f_x = 2x - 4 = 0 \Rightarrow x = 2$. $f_y = 2y + 6 = 0 \Rightarrow y = -3$. The only critical point is $(2, -3)$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find all critical points of $g(x, y) = x^3 - 3x + y^2$.

a. $(1, 0)$ **and** $(-1, 0)$ ✓

b. $(1, 0)$ only

c. $(0, 0)$

d. $(3, 0)$

$g_x = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$. $g_y = 2y = 0 \Rightarrow y = 0$. The critical points are $(1, 0)$ and $(-1, 0)$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the definition of the Hessian discriminant D in the Second Partials Test?

a. $D = f_{xx}f_{yy} - (f_{xy})^2$ ✓

b. $D = f_{xx} + f_{yy}$

c. $D = (f_{xy})^2 - f_{xx}f_{yy}$

d. $D = f_{xx}f_{yy} - f_{xy}$

The discriminant D is the determinant of the Hessian matrix $[[f_{xx}, f_{xy}], [f_{yx}, f_{yy}]]$, which is $D = f_{xx}f_{yy} - (f_{xy})^2$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If at a critical point $D > 0$ and $f_{xx} > 0$, what does the test conclude?

a. f **has a local minimum at the point** ✓

b. f has a local maximum at the point

c. The point is a saddle point

d. The test is inconclusive

When $D > 0$, the surface is locally bowl-shaped; if $f_{xx} > 0$, it curves upward in all directions, establishing a local minimum.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Classify the critical point $(0, 0)$ for $f(x, y) = x^2 - y^2$.

a. **Saddle point** ✓

b. Local minimum

c. Local maximum

d. Inconclusive

$f_x = 2x, f_y = -2y \Rightarrow (0, 0)$. $f_{xx} = 2, f_{yy} = -2, f_{xy} = 0$. $D = (2)(-2) - 0^2 = -4 < 0$. Therefore $(0, 0)$ is a saddle point.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
What does the Second Partials Test conclude when $D = 0$?

a. **The test is inconclusive; higher-order analysis or algebraic examination is needed ✓**

b. The point is definitely a saddle point

c. The point is an absolute extremum

d. The function is not differentiable

When $D = 0$, the test yields no information; the critical point could be a local maximum, local minimum, or saddle point.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
What vector equation governs constrained optimization of $f(x, y)$ subject to constraint $g(x, y) = k$?

a. $\nabla f = \lambda \nabla g$ **(and** $g(x, y) = k$**)** ✓

b. $\nabla f \cdot \nabla g = 0$

c. $\nabla f + \nabla g = \mathbf{0}$

d. $\nabla f = \mathbf{0}$

At a constrained extremum, the level curve of f is tangent to the constraint curve $g = k$, meaning their gradient vectors must be parallel: $\nabla f = \lambda \nabla g$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Maximize $f(x, y) = xy$ subject to the constraint $x + y = 10$ for $x, y > 0$.

a. **Maximum is** 25 **at** $(5, 5)$ ✓

b. Maximum is 24 at $(4, 6)$

c. Maximum is 21 at $(3, 7)$

d. Maximum is 16 at $(2, 8)$

$\nabla f = [y, x]$ and $\nabla g = [1, 1]$. $[y, x] = \lambda[1, 1] \Rightarrow y = \lambda$ and $x = \lambda \Rightarrow x = y$. Using $x + y = 10 \Rightarrow 2x = 10 \Rightarrow x = 5, y = 5$. Maximum $f(5, 5) = 25$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What does the Lagrange multiplier λ represent physically/economically in constrained optimization?

a. The sensitivity / shadow price: the rate of change of the optimal value with respect to the constraint constant k ($\frac{df^*}{dk}$) ✓

b. The curvature of the constraint surface

c. The distance from the origin to the optimal point

d. The second derivative along the constraint

The multiplier $\lambda = \frac{\partial f^*}{\partial k}$ represents the marginal change in the optimized objective value per unit increase in the constraint resource k.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For $f(x, y) = x^3 - 12x + y^2$, classify the critical point at $(-2, 0)$.

a. Saddle point ✓

b. Local minimum

c. Local maximum

d. Inconclusive (D = 0)

$f_x = 3x^2 - 12$, zero at $x = \pm 2$; $f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = 0$, so $D = 12x$. At $x = -2$: $D = -24$: Saddle point.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the x-coordinate of the critical point of $f(x, y) = x^2 - 6x + y^2 - 4y$.

a. 3 ✓

b. -6

c. -3

d. 4

Applying Optimization principles, $f_x = 2x - 6 = 0 \Rightarrow x = 3$. Therefore, the verified correct option is 3.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:

a. Lagrange multiplier ✓

b. eigenvalue

c. gradient

d. curvature

Applying Application of Derivatives principles, λ = Lagrange multiplier. Therefore, the verified correct option is Lagrange multiplier.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

To find candidates for the global max/min of f on $[a, b]$, you check critical points and:

a. **the endpoints** a, b ✓

b. inflection points only

c. only $x = 0$

d. points where f is negative

Global extrema on a closed interval occur at critical points or the endpoints.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The method of Lagrange multipliers finds constrained extrema using multiplier λ. The critical value of λ itself:

a. **is the rate of change of the optimal value w.r.t. the constraint constant** ✓

b. gives the extremum value

c. is always zero

d. equals $|ablaf|$

The Lagrange multiplier = sensitivity (shadow price).

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Optimization with constraint (Lagrange): extremise f s.t. $g = 0$ via $ablaf = \lambda ablag$. Number of equations for n vars, 1 constraint:

a. $n + 1$ ✓

b. n

c. $n - 1$

d. $2n$

n gradient equations + 1 constraint = $n + 1$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use spherical coordinates to find the volume of the upper half of the ball $x^2 + y^2 + z^2 \le 4$.

a. $\frac{16\pi}{3}$ ✓

b. $\frac{32\pi}{3}$

c. $\frac{8\pi}{3}$

d. 4π

$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$; integrating over the region gives $\frac{1}{2} \cdot \frac{4}{3}\pi(2)^3 = \frac{16\pi}{3}$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{32\pi}{3}$ ✓

d. $\frac{16\pi}{3}$

$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$; integrating over the region gives $\frac{1}{8} \cdot \frac{4}{3}\pi(4)^3 = \frac{32\pi}{3}$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Double Integrals

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle -y, x \rangle$ along the circle $x = 2\cos(t),\ y = 2\sin(t),\ 0 \le t \le \pi$.

a. -4π

b. 4π ✓

c. 8π

d. 5π

Substitute the parametrization: $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 4$. Integrating over $[0, \pi]$ gives 4π.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, -x, 1 \rangle$ along the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$, $0 \le t \le 2\pi$.

a. π

b. 1

c. 0 ✓

d. -1

Substitute the parametrization: $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0$. Integrating over $[0, 2\pi]$ gives 0.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the sphere $x^2 + y^2 + z^2 = 9$.

a. $\frac{2916\pi}{5}$ ✓

b. $\frac{5832\pi}{5}$

c. $\frac{1458\pi}{5}$

d. $-\frac{2916\pi}{5}$

$\nabla \cdot \mathbf{F} = 3x^2 + 3y^2 + 3z^2$. The flux equals $\iiint_E \nabla \cdot \mathbf{F} \, dV = \frac{2916\pi}{5}$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Double Integrals

Use Green's theorem to evaluate $\oint_C -y^3\,dx + x^3\,dy$, where C is the circle $x^2 + y^2 = 4$, oriented counterclockwise.

a. -24π

b. 48π

c. 12π

d. 24π ✓

$\oint_C P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA$ with $Q_x - P_y = 3x^2 + 3y^2$. In polar coordinates this equals 24π.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Use Green's theorem to evaluate $\oint_C y\,dx + 3x\,dy$, where C is the boundary of the rectangle $[0,1] \times [0,1]$, oriented counterclockwise.

a. -2

b. 4

c. 1

d. 2 ✓

$\oint_C P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA$ with $Q_x - P_y = 2$. $\int_0^1 \int_0^1 2\,dx\,dy \;=\; 2$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Evaluate $\int_C x^2\,ds$ where C is the segment from $(0,\ 0)$ to $(5,\ 12)$.

a. $\frac{25}{3}$

b. $\frac{4225}{3}$

c. $\frac{328}{3}$

d. $\frac{325}{3}$ ✓

$\mathbf{r}(t) = \langle 5t, 12t \rangle,\ ds = 13\,dt$. $\int_0^1 325t^2\,dt \;=\; \frac{325}{3}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Double Integrals

Use Green's theorem to find the area enclosed by the astroid $x = 1\cos^3 t,\ y = 1\sin^3 t$.

a. $\frac{3\pi}{8}$ ✓

b. π

c. $\frac{3\pi}{4}$

d. $\frac{3\pi}{16}$

$x\,dy - y\,dx = 3\sin^2 t\cos^2 t\,dt$. Then $A = \frac{1}{2}\int_0^{2\pi} 3\sin^2 t\cos^2 t\,dt = \frac{3\pi}{8}$.

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